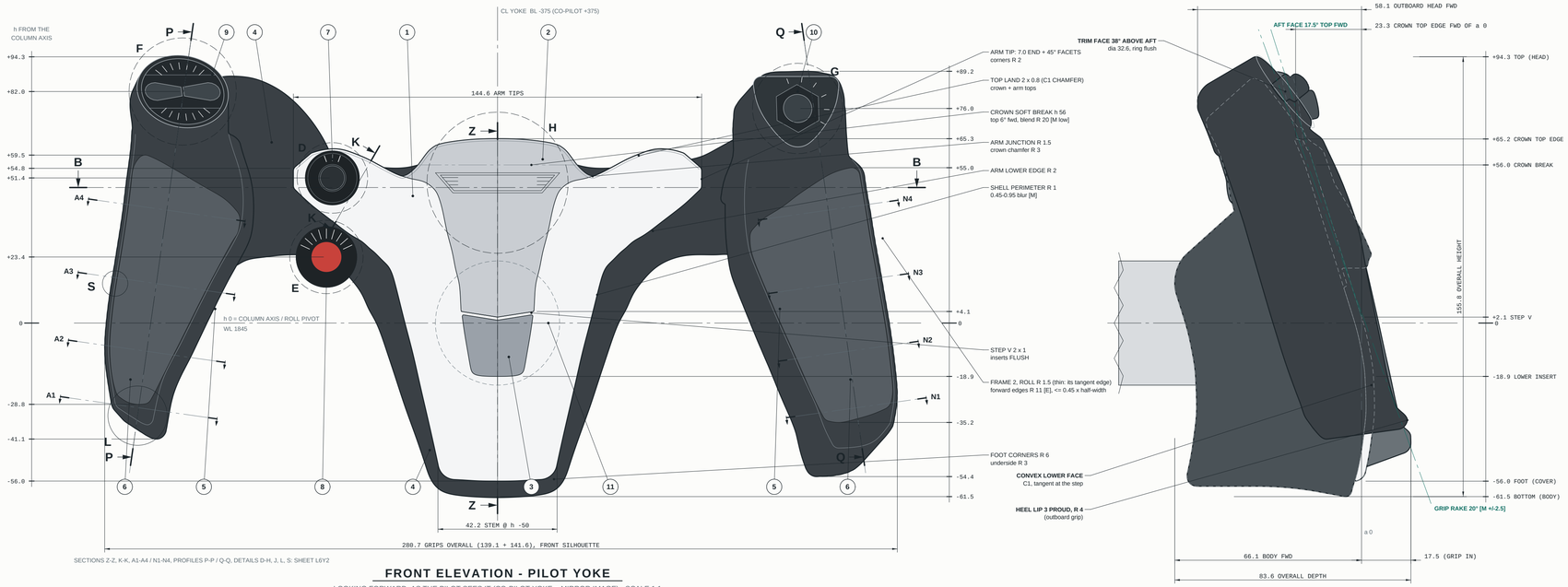


SECTIONS Z-Z, K-K, A1-A4 / N1-N4, PROFILES P-P / Q-Q, DETAILS D-H, J, L, S: SHEET L6Y2

FRONT ELEVATION - PILOT YOKE

LOOKING FORWARD, AS THE PILOT SEES IT (CO-PILOT YOKE = MIRROR IMAGE) - SCALE 1:1

SIDE ELEVATION

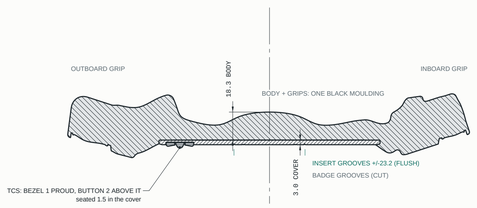
FROM OUTBOARD, NOSE LEFT, NEUTRAL PITCH: a 0 = X 3700; COLUMN BROKEN (L6Y2 DETAIL J) - 1:1

GRIP PLANES [M], 6 VIEWS INCL. P1046409: RAKE 20 (+/-2.5), a0 -3.75 (+/-3), DEPTH 27 (+/-4), WALLS 0.6 / 0; TOE 18 WEAK (+/-8, OPEN POINT 2)

BODY (HIDDEN): TOP 42° FROM THE HORIZONTAL, TOP-TO-FRONT R 6, DEPTH 60 [E], FOOT R 5 [E] (L6Y2 Z-Z)

VIEW FROM ABOVE

FORWARD UP, PILOT YOKE - SCALE 1:1; GRIP PLANES ON 6 VIEWS (L6Y2 PHOTO CHECK); TOE WEAK (OPEN POINT 2)



SECTION B-B

h 48 (BADGE / TCS), EXACT CUT OF THE SOLIDS, FROM ABOVE - SCALE 1:2

REVISIONS - PC12-L6Y

REV	DATE	DESCRIPTION
A	2026-10-05	first issue (owner 2026-10-05: 'the yoke isn't quite right especially if you look at the details'): yoke re-measured on 8 PRO photos; YOKE_DESIGN rev I replaces the rev H outline
B	2026-10-05	rev J (judges r1): the yoke as solids (model/yoke_geom.py) -- every view / section exact; curve law + corner tables; one body with an outboard web and an inboard LOW NECK (open notch under the head); pads with diagonal tops in a raised 2 mm frame, rolled heel lip; grips refitted per grip (toe 18 / 18) on 4 pad views; ROUND trim face and SHIELD facet heads, triangulated switch centres; crown / arm tops 1.7 lower (landmark-pose frame), top land, crown crease; body top falling forward 35 deg; inserts flush; AFCS round boss, hex button collar, mirrored rocker lobes; sheet L6Y2 added (sections, heads, switches, column, photo check)
C	2026-10-05	rev K (judges r2): exact 2-D distance fields (no raster), Stage-3 surface route + acceptance checks; grip walls fore-aft (outboard 0.6) -- slot-side and heel silhouettes on the curated photo data; forward-edge R capped, PCHIP depth, heel lip R 4, pad groove + bead built; body top 42 deg from horizontal, top-to-front R 8; the web held under the trim face; AFCS boss, TCS, rocker, hex button as solids; TCS button proud; crown soft break R 20; grip planes JOINT-FITTED to 7 views (5 confirm the table +/-3 deg / +/-3-5 mm; the AOPA photo disagrees: open); outboard heel corner pulled in; ordinate dimensions, cutting planes, detail circles, all values computed
D	2026-10-05	rev L (judges r3): P1046409 posed (two-view camera) -- grip planes re-fitted on 6 views, depth 27 from its slot wall (no bottom taper), PROVISIONAL removed, the AOPA photo dropped; the web runs into the trim head's rim (no fin) on the face's plane; head zone of named surfaces and tabled rounds; body top = a plane through the outline's top edge, rounds in order; C2 curve law; U column recess [M]; forward-face switches unverified (PTT expected); knee sweep with the inboard grip's lowest band restored (rev K re-swept); acceptance clean: the head zone's draft starts C1, inboard cut R 1.5, foot R 5 under a 4 mm draft cap, unit-gradient top / aft

ITEMS & FINISHES (render/lookdev names; 'new' = to add)

#	PART	MATERIAL	FINISH / COLOUR [SOURCE]
1	WHITE COVER (Y SHELL)	yoke_white	gloss clear-coat white, neutral-cool (0.77 / 0.78 / 0.80), fine pearl; R 1 edges, top land [M]
2	UPPER INSERT = CROWN, BADGE	yoke_silver (new)	satin aluminium 0.62, metallic 0.85, rough 0.30; flush, 0.6 x 0.5 grooves [M]
3	LOWER INSERT	yoke_silver_dark (new)	smoke silver 0.30, metallic 0.85, rough 0.34; flush [M]
4	BLACK BODY (RIM, WEB, NECK)	grip_black	satin black, rough 0.42: crisp edge highlights; R 2 aft edge [M]
5	GRIP (2) WITH FRAME, HEEL LIP	grip_black	satin black; 2.0 frame R 1.5, heel lip 3 proud R 4 [M]
6	LEATHER PAD (2)	leather_black	charcoal 0.05, rough 0.58, 1.1 pebble grain, ~3 creases, 2.5 below the frame, groove 1.0 x 0.5 + bead 0.4 at its edge [M / E]
7	TCS BUTTON	gloss_black / black	gloss black bezel 1 proud [E], satin domed button 2 proud of it [M low]
8	AFCS DISCONNECT	button_red (new) / black	red cap 0.70 / 0.035 / 0.04, clear coat 0.3, NOT emissive, on a black boss [M]
9	PITCH TRIM ON ROUND FACE	black / legend_white	satin black face, ring + rocker, tick legends 0.80 [M]
10	PUSH BUTTON ON SHIELD	black	satin black hex collar, smooth cap, centre dimple [M]
11	COLUMN	metal_dark	satin dark metal, dia 44 [E] (never seen)
12	COLUMN SURROUND (PANEL)	pedestal_gunmetal	satin dark grey J band 23 wide + dished recess [M / E]

MAIN DIMENSIONS (read from the table / the solids)

DIMENSION (mm, YOKE-LOCAL)	VALUE	SOURCE
shell height / about h 0	121.3 / -56.0..+65.3	[M] rev J landmark-pose frame
arm tips / stem at h 0 / at h -50	144.6 / 65.6 / 42.2	[M] sigma 1.5
crown above the arm junctions	13.3	[M] parting grooves
step / lower insert bottom	h 2.1 (V) / -18.9	[M]
black rim outboard of the stem at h 10 / 0 / -50 / under the foot	12.6 / 10.1 / 4.5 / 5.8	[M] body - shell outlines
aft face above the step	17.5° top fwd	[M] bundle adj., +/-4
body depth / top / cover	60 / 42° from horiz. / 3	[E] / [M lower bound] / [E]
heads top: FRONT SILHOUETTE out / in	94.3 / 89.2	[derived: solids from M / E]
grip bottoms (front silhouette) out / in	-41.1 / -54.4	[derived: solids, rake / toe M, depth M / E]
grips overall: front silhouette (out + in)	280.7 (139.1 + 141.6)	[derived: solids]
CONSTRUCTION G (prism directrix): head tops / span	97.4 / 90.1; 281.3	[M] table; NOT the part (head solids trim it)
rake / toe-out / a0 @ h 14: out; in	20° / 18° / -3.75; 20° / 18° / -6	[M] 6 views incl. P1046409: +/-2.5° / toe weak / +/-3
walls (0 = aft normal, 1 = fore-aft) out / in	0.6 / 0	[M] +/-0.2
grip depth (normal) bottom / mid / head, out; in	27.5 / 27 / 27; 27.5 / 27 / 27	[M] P1046409 slot wall +/-4; in = out [E]
pad out: top / slot corner / bottom	59.5 / (91, 38) / -28.8	[M] diagonal top 31°
pad in: top / slot corner / bottom	55.0 / (89.4, 32) / -35.2	[M] diagonal top 33°
frame / leather below it / heel lip	2 / 2.5 / 3 proud R 4	[M]
trim face centre / dia / up, turn	(111.5, 82) / 32.6 / 38°, 10°	[M] triangulated, 7 views
shield facet / button centre	31 x 25.5 / (106.3, 76)	[M] (inboard: -u)
TCS / AFCS cap / boss	(58.5, 51.4) 19.2 / 10.8 x 5.5 / 21.6	[M]
column dia / axis	44 [E] / h 0 = WL 1845	kept; surround reads 1821
pitch travel / roll	+/-90 [E] / +/-70 [E]	unchanged (YOKE)

KNEES vs YOKE (L6B), REV H > I > J > K (RE-SWEPT) > L - CRITERIA UNCHANGED

OCCUPANT	LEVEL mm	ROLL+/-70 mm	CONTACT deg	KEEP 25 deg	RUDDER mm
95th male	38 > 35 > 40 > 25 > 24	-19 > -18 > -19 > -19 > -19	17.5 > 17.5 > 15.0 > 12.5 > 12.5	5.0 > 2.5 > 5.0 > 0.0 > -1.0	-11 > -13 > -9 > -22 > -24
50th male	50 > 48 > 52 > 38 > 36	-13 > -14 > -14 > -14 > -14	25.0 > 25.0 > 22.5 > 22.5 > 22.5	10.0 > 10.0 > 10.0 > 5.0 > 5.0	5 > 3 > 7 > -7 > -8
5th male	51 > 48 > 51 > 37 > 35	-21 > -21 > -22 > -22 > -21	22.5 > 22.5 > 22.5 > 17.5 > 17.5	10.0 > 10.0 > 10.0 > 5.0 > 5.0	20 > 18 > 22 > 8 > 7
5th female	76 > 78 > 81 > 67 > 65	5 > 0 > 0 > 0 > 0	87.5 > 87.5 > 87.5 > 87.5 > 87.5	27.5 > 25.0 > 25.0 > 20.0 > 20.0	55 > 53 > 56 > 42 > 40

NOTES

1. Yoke-local mm: u outboard (pilot -y), h up from the hub centre = the column axis / roll pivot (WL 1845, BL -/+375), a aft of the aft face at the insert step (X 3700 at neutral pitch). Drawn from model/interior.py YOKE_DESIGN rev I, through the solids of model/yoke_geom.py; L6 / L6B take the same solids (yoke_outline_yz, yoke_sweep_pieces).
2. CURVE LAW (table curve_law, rev L): every outline is a C2 cubic spline on centripetal knots through its samples (curvature continuous at every sample that is not a corner), sampled every 0.25; the samples in the *_corners tables are corners (R 0 sharp, R > 0 a tangent fillet); the lower aft face is a C1 Hermite tangent to the 17.5° plane. The solids use the EXACT distance to those polylines (no raster); views are exact silhouettes, sections exact cuts.
3. SURFACES: the cover's face (the 17.5° plane, the lower Hermite, the crown bevel) trimmed by its outline; the body: sides / bottom = its outline without the top edge, top = a plane 42° from the horizontal through the outline's top edge (body_top), forward face, rounds in order R 2 / R 2 / R 8 (R 5 at the foot, the draft's offset capped at 4 there); the web's aft face = the trim face's plane 3.5 below it (web); each grip = its prism (outline G, a construction curve, on the raked / toed plane) + its HEAD (head_zone: the prism trimmed .1 inside the head's shaping surface drafted outward below the plane height 70 / 61 (a C1 start), edge R 1.5; the head solid = round face on its flared cylinder / shield facet + fall-off, x chamfer R 3, x forward face R 3, x walls R 2, floor R 3; joined R 1); body + grips one moulding, joined by a true round R 3. Every edge is a tabled round or a tabled crease (L6Y2 CORNERS & RADII).
4. STAGE-3 SURFACE ROUTE: each named surface a grid_surface over its parametrisation (planes, the face Hermite, the head cylinder about the face normal) trimmed by cad.mesh.trim with these exact 2-D fields; each tabled round a sweep_profile of its radius; the head-zone trimming surface is a draft from the head surface's offset curve; vertex normals from the analytic surfaces. The solids stay the inside / outside oracle. Acceptance (model/yoke_checks.py, L6Y2): vertex normals within 2° at the 95th percentile on 9 patches incl. the heads, web, tops and forward faces; no untabled section cusps (grips to the head tops, necks, h planes); one piece per side in every h section above 62; nothing above the trim face / shield.
5. White moulding symmetric; the black body and the grips are NOT: an outboard web running into the trim head's rim (10-11 o'clock), an inboard LOW NECK with an open notch under the inboard head; switches outboard (TCS, AFCS, trim), the MAP LIGHTS push button on the inboard head; co-pilot yoke = mirror image.
6. NO MARKINGS: the badge is an abstract winged plate, every legend a row of ticks.
7. Not modelled: the rev H column bellows (none in 8 photos), the PC-24's grip tab (not on the PRO). The grips' FORWARD (finger-side) faces are UNVERIFIED -- no photo shows them; AOPA (2025-03-17) lists push-to-talk and checklist among the yoke's one-hand functions, and only trim, TCS, AFCS DISC and MAP LIGHTS are seen: the PTT (and a checklist switch) are expected there (open point 7).
8. GRIPS' 3-D LIE (L6Y2 photo check): rake / toe / a0 / walls / depth fitted per grip to the curated silhouettes of 6 views -- P1046408 L / R, P1046410, Mexico L / R and P1046409 (posed this round: a two-view camera with panel_408, the co-pilot yoke 25° from inboard) -- with the front outline held: rake 20 +/-2.5, a0 -3.75 +/-3, walls 0.6 / 0 +/-0.2, depth 27 +/-4 (P1046409's slot wall; no bottom taper); the toe is weakly determined by views from inboard only (open point 2). The AOPA photo is not used (one radial term, pose degenerate along the line of sight).
9. Knee checks on the rev L solids: the 95th clears the yoke by 23.8 mm wings level (criterion 25: MISSED by 1.2; the lowest band of the inboard grip, whose forward face now runs the measured 27.5 deep to the bottom); roll to contact 12.5°, keeping 25 mm to 0.0° -- owner decision (open point 1 of L6B; criteria not loosened). K column = rev K re-swept: as issued (37 mm, 15°) the sweep had dropped that band (a doubled vertex in the band clip, fixed).

OPEN POINTS FOR THE OWNER

1. Fore-aft station: EVERY posed photo puts the aft face forward of the drawn neutral -- P1046408 -60 / -55, P1046410 -65, P1046409 -37 (s/n 3001, gust lock on: 'CONTROLS LOCKED'), Mexico -38 / -34 and the AOPA photo -14 / -26 (no lock placard) [+/-20-30 each along the line of sight]. Unlocked, a parked yoke rests forward: the elevator droops trailing-edge down under its weight [E: physics, no PC-12 source]. Option: neutral 45 further forward (hub_dx 140 -> 95): the 95th's contact roll 12.5° -> 12.5°, wings-level clearance 23.8 -> 23.8 mm (L6B sweep: the knees lie inside the +/-90 pitch-swept slab either way, so the station does not move the knee rows). Kept 140 (owner).
2. Grip toe: all six usable views look from INBOARD (15-27°), which fixes rake, a0, walls and depth but leaves the toe +/-8 (silhouette cost +5 % at 18 vs 7; the pads flat 0-26); the one view from outboard (AOPA, camera caveat) favours 18-26. Kept 18; a view from outboard of a PRO yoke would settle it.
3. Column: the surround's joint line reads WL 1821 +/-10 (25 under the pivot, kept); J band or a full collar round the column (L6Y2 detail J); the recess a U 114 wide to h -72 [M]; push travel 90 [E].
4. Shell skew: posed on the symmetric landmarks the stem sits ~1.6 inboard and the arm tips ~1.5 outboard of the inserts' centreline in all 7 views (TCS / AFCS u fit within 0.7): carry the <= 2 mm skew or keep it symmetric (kept).
5. Knees: the 95th clears 23.8 wings level (criterion 25) and meets a grip from 12.5° of roll (rev K re-swept 25.3 / 12.5°): owner decision, criteria unchanged.
6. Crown soft break (R 20 [E]) vs a crease; hidden: body depth 60, the necks' depth 30, rocker / cap heights.
7. Forward-face switches (PTT, checklist): expected from the AOPA function list, never seen -- a photo of a grip's finger side would add them.

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - VIEWS

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y	D	AS SHOWN	A1 · 1 OF 2	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-10-05

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION